

SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-2023



Program/s: MBA Tech/B.Tech Data Science

Year: II Semester: IV

Stream/s ; Data Science

Subject: Statistical Structures in Data and Inference

Time: 3 hrs (10:00 am to 1:00 pm)

Date: 25 / 04 / 2023

No. of Pages: 5

Marks: 100

Final Examination / Re - Examination.

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1 ; SO- ;1 BL-2	a.	Illustrate the assumption of ANOVA	[5]
CO-1 ; SO- ;1 BL-2	b.	Indicate the difference between Ridge and lasso regression model?	[5]
CO-2 ; SO- ;1 BL-4	c.	How to estimate the likelihood function for binomial distribution for parameter estimation?	[5]
CO-1 ; SO- ;3 BL-2	d.	Explain Potential Problems of Regression	[5]
Q2			[20]
CO-3; SO- ;3 BL-3	A	Produce python code to calculate mean and standard deviation through Maximum likelihood estimation in the below data set refer as df. <i>Python code:</i> import numpy as np from scipy.optimize import minimize from scipy.stats import norm np.random.seed(123) N=500	[10]

		$\mu = 5$ $\sigma = 9$ $df = np.random.normal(loc = \mu, scale = \sigma, size = (N,))$																									
CO-2; SO- ;1 BL-1	B	Explain how the parameters of the normal distribution is calculated through maximum likelihood estimation	[10]																								
Q3			[20]																								
CO-3; SO- ;2,5 BL-3	A.I	Before and after as objects containing the scores of python training in a company. Find out whether the training program is effective or not. <i>Python code:</i> <pre>from scipy import stats after = [12.2, 14.6, 13.4, 11.2, 12.7, 10.4, 15.8, 13.9, 9.5, 14.2] before = [13.5, 15.2, 13.6, 12.8, 13.7, 11.3, 16.5, 13.4, 8.7, 14.6]</pre> Produce python code and analyze the output	[5]																								
CO-3; SO- ;2,5 BL-3	A.II	Below data hold plant heights from each sample <i>Python code:</i> <pre>G1 <- [8, 8, 9, 9, 9, 11, 12, 13, 13, 14, 15, 19] G2<- [11, 12, 13, 13, 14, 14, 14, 15, 16, 18, 18, 19]</pre> Find out whether the mean height of both groups are same or not. Produce python code and analyze the output	[5]																								
CO-1; SO- ;2,5 BL-4	B	Reena believes that 50% of first-time brides in the India are younger than their grooms. She performs a hypothesis test to determine if the percentage is the same or different from 50%. Reena samples 100 first-time brides and 53 reply that they are younger than their grooms. Analyze whether what Reena believes is true or false at 1% level of significance.	[10]																								
Q4			[20]																								
CO-2; SO- ;2,12 BL-4	A	Four brands of flashlight batteries are to be compared by testing each brand in five flashlights. Twenty flashlights are randomly selected and divided randomly into four groups of five flashlights each. Then each group of flashlights uses a different brand of battery. The lifetimes of the batteries, to the nearest hour, are as follows. <table border="1"> <thead> <tr> <th>Brand A</th><th>Brand B</th><th>Brand C</th><th>Brand D</th></tr> </thead> <tbody> <tr> <td>42</td><td>28</td><td>24</td><td>20</td></tr> <tr> <td>30</td><td>36</td><td>36</td><td>32</td></tr> <tr> <td>39</td><td>31</td><td>28</td><td>38</td></tr> <tr> <td>28</td><td>32</td><td>28</td><td>28</td></tr> <tr> <td>29</td><td>27</td><td>33</td><td>25</td></tr> </tbody> </table>	Brand A	Brand B	Brand C	Brand D	42	28	24	20	30	36	36	32	39	31	28	38	28	32	28	28	29	27	33	25	[10]
Brand A	Brand B	Brand C	Brand D																								
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29	27	33	25																								

		Preliminary data analyses indicate that the independent samples come from normal populations with equal standard deviations. At the 5% significance level, does there appear to be a difference in mean lifetime among the four brands of batteries?	
CO-3; SO-; BL-3	B	A new type of corn seed has been developed and a team of agronomists wanted to determine whether there was a significant difference between the types of soils that they are planted in (loam, sandy, salty, clay) based on the yield of the crop, amount of water required and amount of herbicide needed. four fields of each type were chosen for the analysis. Based on the data determine whether there is a significant difference between the results for each type of soil condition. <i>Python code:</i> <pre> from statsmodels.formula.api import ols from statsmodels.base.model import Model from statsmodels.stats.multicomp import pairwise_tukeyhsd import pandas as pd import numpy as np import statsmodels.api as sm from statsmodels.multivariate.manova import MANOVA df=pd.DataFrame({'Soil':np.repeat(['loam','sandy','salty','clay'],4), 'Yield':[76.7,60.5,96.1,88.1,66.9,55.4,50.5,64.1,62.8,45,47.8,75.6,46.3,61.5,62.9,49.3], 'Water':[29.5,32.1,40.7,45.1,23.9,29.1,18,14.5,25.9,15.9,36.1,27.7,31.8,16.8,25.8,39.4], 'Herbicide':[7.5,6.3,4.2,4.9,1.1,5,4.8,3.7,2.9,1.2,4.1,6.3,7.4,1.9,2.4,5.2]}) </pre>	[10]
Q5			[20]
CO-3; SO-; BL-3	A	Briefly explain multiple regression. Find out the best multiple regression model taking mpg as a dependent variable. <i>Python Code:</i> <pre> import statsmodels.api as sm import statsmodels.formula.api as smf mtcars = sm.datasets.get_rdataset("mtcars", "datasets", cache=True).data mtcars </pre> Produce python code and analyze the output.	[10]
	B	State the assumption of linear regression	[10]
Q6			[20]
	A	<i>Python Code:</i>	[10]

CO-3; SO-; BL-3		<pre>import pandas as pd df = pd.DataFrame({'x':[1,2,3,5,6,7,8,9,10,12,13,14,15,16,18,19,21,22], 'y':[100,90,80,60,60,55,60,65,70,70,75,76,78,79,90,99,99,100]})</pre> Build a best polynomial regression model on the above data set. Produce python code and analyze the output.																																								
CO-3; SO-; BL-3	B	<i>Python Code:</i> <pre>from sklearn.metrics import confusion_matrix,classification_report import statsmodels.api as sm from sklearn import linear_model import pandas as pd df3=pd.DataFrame({'x1':[2.78,2.44,2.09,2.14,2.72,2.65,4.92,4.37,4.96,4.52,4.69 ,3.88], 'x2':[78,74,79,4,72,65,9,7,6,52,9,8], 'y':[0,0,0,0,0,0,1,1,1,1,1,1]})</pre> Briefly explain logistic regression and build a logistic regression model on the above data set. Produce python code, analyze the output and confusion matrix	[10]																																							
Q7			[20]																																							
CO-2; SO-; BL-4	A	The table below shows the hours of relief provided by two analgesic drugs in 12 patients suffering from arthritis. Is there any evidence that one drug provides longer relief than the other? (*Use sign test) <table><tr><td>Case</td><td>Drug A</td><td>Drug B</td></tr><tr><td>1</td><td>2</td><td>3.5</td></tr><tr><td>2</td><td>3.6</td><td>5.7</td></tr><tr><td>3</td><td>2.6</td><td>2.9</td></tr><tr><td>4</td><td>2.6</td><td>2.4</td></tr><tr><td>5</td><td>7.3</td><td>9.9</td></tr><tr><td>6</td><td>3.4</td><td>3.3</td></tr><tr><td>7</td><td>14.9</td><td>16.7</td></tr><tr><td>8</td><td>6.6</td><td>6</td></tr><tr><td>9</td><td>2.3</td><td>3.8</td></tr><tr><td>10</td><td>2</td><td>4</td></tr><tr><td>11</td><td>6.8</td><td>9.1</td></tr><tr><td>12</td><td>8.5</td><td>20.9</td></tr></table>	Case	Drug A	Drug B	1	2	3.5	2	3.6	5.7	3	2.6	2.9	4	2.6	2.4	5	7.3	9.9	6	3.4	3.3	7	14.9	16.7	8	6.6	6	9	2.3	3.8	10	2	4	11	6.8	9.1	12	8.5	20.9	[10]
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CO-3; SO-; BL-4	B	The Data “Salary” contains 2008-09 nine-month academic salary for Assistant Professors, Associate Professors and Professors in a college in the U.S. First, build a regression model considering salary as dependent variable and others as	[10]																																							

		independent variable. Secondly build a regression model considering qualitative variable as only independent variables and interpret the results.	
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